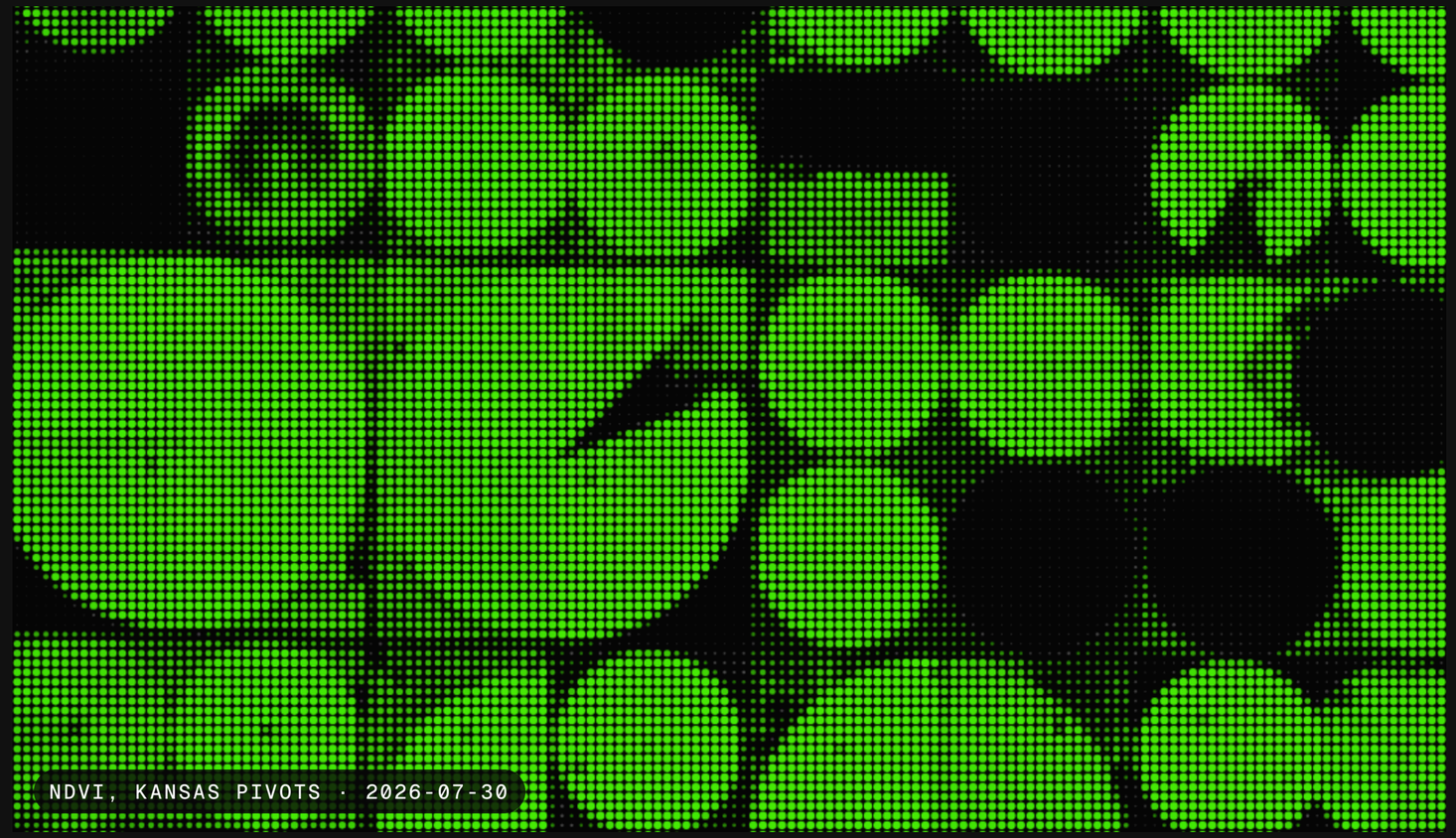
HRT — HORTUS · PLANT VISION AND AUTOMATION

hortus

Seeing what plants need.

AI vision for plant analytics, and a five-tier path from a phone to an autonomous room.



A photo is the cheapest sensor on a farm. It is also the easiest to over-trust.

Lab accuracy is not field accuracy.

99.35%

PLANTVILLAGE, HELD-OUT

The same network on photos taken elsewhere: 31.4%.

49.0%

FROM 8 BACKGROUND PIXELS

PlantVillage labels leak through the backdrop, not the leaf.

56.9%

GPT-4O ZERO-SHOT, APPLE LEAVES

98.1% after fine-tuning on the same images.

MOHANTY ET AL. (2016); NOYAN (2022); ROUMELIOTIS ET AL. (2025)

What colour can and cannot tell.

It can

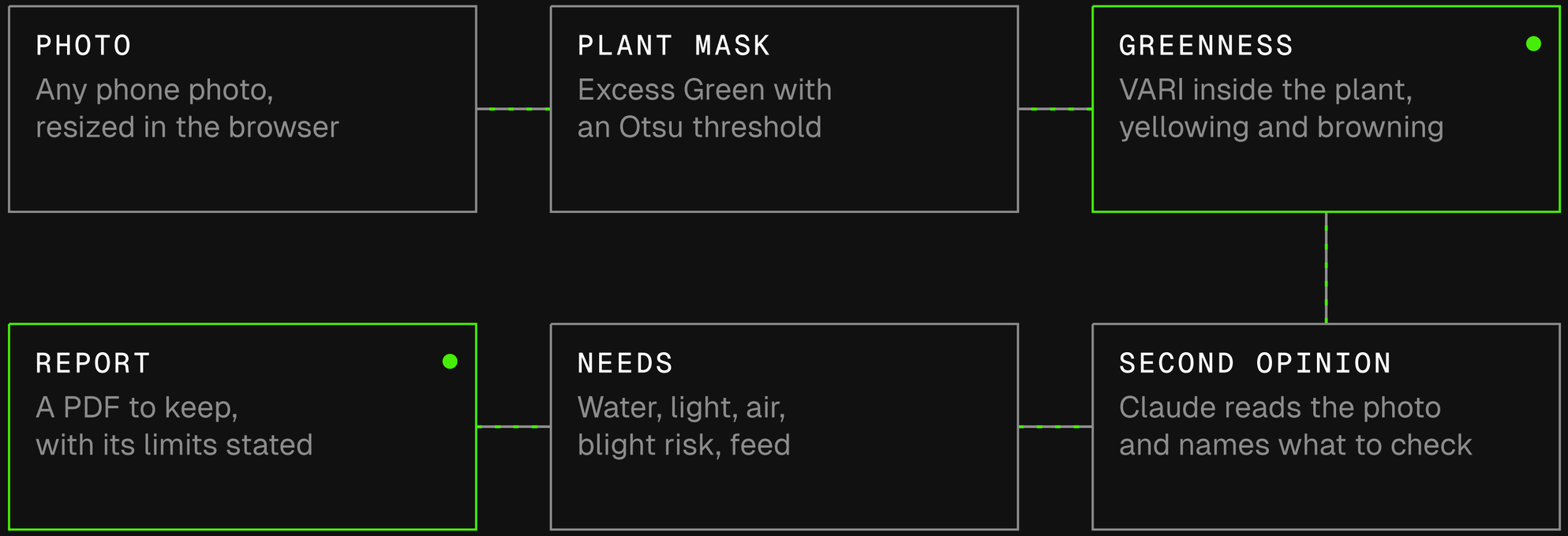
Separate plant from soil (Excess Green). Measure greenness inside the plant (VARI). Map where tissue turns yellow or brown. Track change between photos of the same plant.

It cannot

Tell nitrogen shortage from a virus or from drought. Read a disease from a single image with certainty. Replace a field check, a lab test or an agronomist.

WOEBBECKE ET AL. (1995); GITELSON ET AL. (2002); OTSU (1979)

The analyser, step by step.



THE ANALYSER RUNS IN THE BROWSER; THE SECOND OPINION IS OPTIONAL AND ASKS THE VIEWER FIRST

Five tiers, from a phone to an autonomous room.

TIER 0

Look

Phone photo, AI vision, a log

TIER 1

Sense

Soil moisture, air temperature and humidity

TIER 2

Control

Closed loops for feed, pH, climate and CO_2

TIER 3

Predict

Weather, satellite and crop models

TIER 4

Autonomy

Model-predictive control and robotics

Tier 1: sense the root zone and the air.

TIER 0	TIER 1	TIER 2	TIER 3	TIER 4
Look	**Sense**	**Control**	**Predict**	**Autonomy**
Phone photo, AI vision, a log	Soil moisture, air temperature and humidity	Closed loops for feed, pH, climate and CO_2	Weather, satellite and crop models	Model-predictive control and robotics

HortusRoom S

Waters from a moisture reading. Watering 10 s to 60 min, pause 1 min to 10 h, pump up to 1,500 W.

HortusRoom A

Heat and humidity on two channels, up to 1,500 W each, −40 to +80 °C.

Tier 2: close the loops.

TIER 0	TIER 1	TIER 2	TIER 3	TIER 4
Look	**Sense**	**Control**	**Predict**	**Autonomy**
Phone photo, AI vision, a log	Soil moisture, air temperature and humidity	Closed loops for feed, pH, climate and CO_2	Weather, satellite and crop models	Model-predictive control and robotics

HortusPilot N and F

Feed to the set EC, then correct pH. EC 0.1–5.0 mS, tanks 10–2,000 L; the F model watches 15 sensors and 26 alarms.

HortusPilot pH · HortusRoom C

pH 3.0–12.0 under a PI controller. CO_2 held between 400 and 2,000 ppm, raised only while plants photosynthesise.

Tier 3: predict the need before it shows.

TIER 0	TIER 1	TIER 2	TIER 3	TIER 4
Look	**Sense**	**Control**	**Predict**	**Autonomy**
Phone photo, AI vision, a log	Soil moisture, air temperature and humidity	Closed loops for feed, pH, climate and CO_2	Weather, satellite and crop models	Model-predictive control and robotics

WATER	$ET_0 \times Kc$	FAO-56 PENMAN-MONTEITH WITH CROP COEFFICIENTS	STRONG
LIGHT	$DLI = PPFD \times h \times 0.0036$	LEAFY GREENS ≥ 12, FRUITING CROPS ≥ 15 MOL M^{-2} D^{-1}	MODERATE
BLIGHT	Hutton criteria	TWO DAYS ≥ 10 °C MINIMUM WITH ≥ 6 H AT RH ≥ 90%	MODERATE

ALLEN ET AL. (1998) FAO-56; MSU EXTENSION DLI GUIDANCE; AHDB HUTTON CRITERIA (2017)

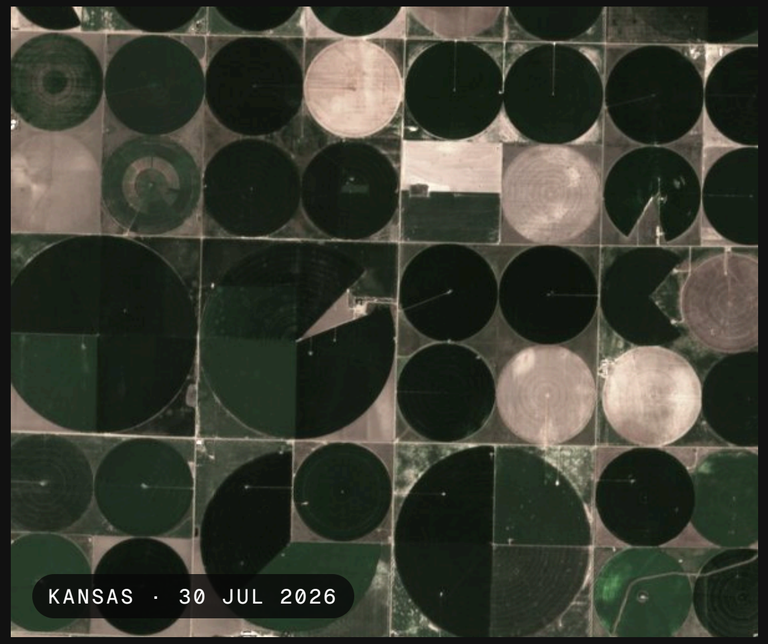
Tier 4: autonomy, with people auditing it.

TIER 0	TIER 1	TIER 2	TIER 3	TIER 4
Look	**Sense**	**Control**	**Predict**	**Autonomy**
Phone photo, AI vision, a log	Soil moisture, air temperature and humidity	Closed loops for feed, pH, climate and CO_2	Weather, satellite and crop models	Model-predictive control and robotics

In the first Autonomous Greenhouse Challenge, five teams grew cucumbers remotely with AI; one out-produced the growers' manual reference. Evidence across crops and sites is still thin.

HEMMING ET AL. (2019), SENSORS 19(8):1807

Three fields, captured in 2026.



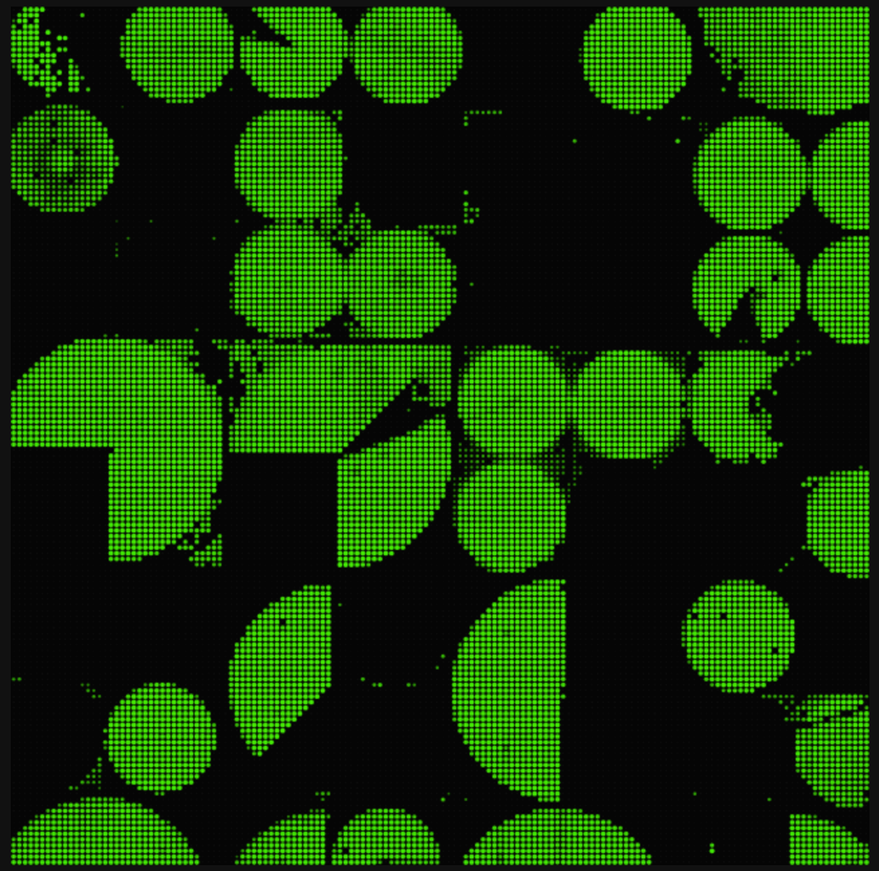


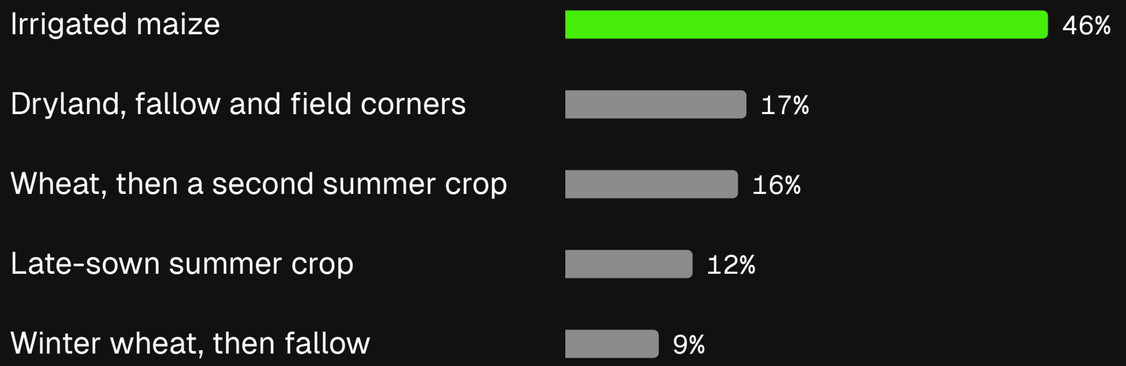


Each window is 6 × 6 km at 10 m: 39, 47 and 53 usable scenes after cloud masking.

DATA: COPERNICUS SENTINEL-2 L2A (EARTH SEARCH, AWS) · NASA POWER (MERRA-2, FLASHFLUX, IMERG)

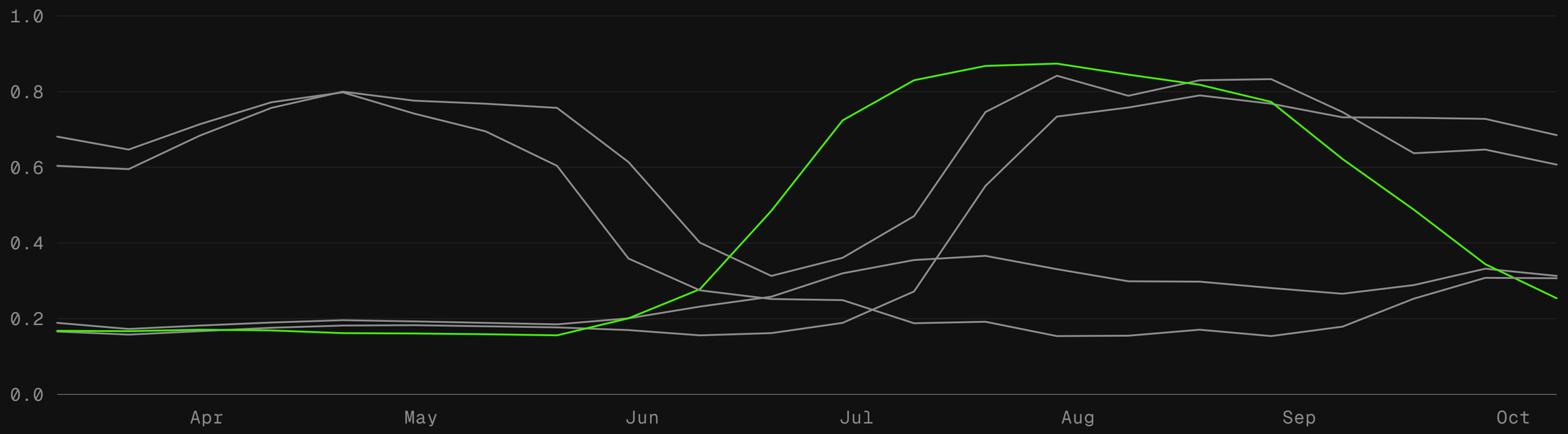
Kansas: five crop calendars in 36 km².





DATA: COPERNICUS SENTINEL-2 L2A (EARTH SEARCH, AWS) · NASA POWER (MERRA-2, FLASHFLUX, IMERG) · K-MEANS ON 10-DAY NDVI CURVES

Kansas: the season, curve by curve.



Wheat greens in spring and is cut in June; irrigated maize peaks in late July; a second crop follows the wheat.

DATA: COPERNICUS SENTINEL-2 L2A (EARTH SEARCH, AWS) · NASA POWER (MERRA-2, FLASHFLUX, IMERG)

Kansas: what the maize needed in a hot summer.

735–936 mm

CROP WATER USE

05-27 to 10-01

436 mm

RAIN (IMERG)

Same window

520–714 mm

NET IRRIGATION NEED

10-day budgets, 80% rain efficiency

+2.9 °C

SUMMER HEAT

June–August mean maximum 36.5 °C against 33.6 °C in 1991–2020; level with 2012

FAO-56 WITH AN NDVI-SCALED CROP COEFFICIENT; RANGE SPANS PENMAN-MONTEITH AND HARGREAVES

Mkushi: in the dry season, water only comes from the pivot.



26%
WINDOW UNDER PIVOTS

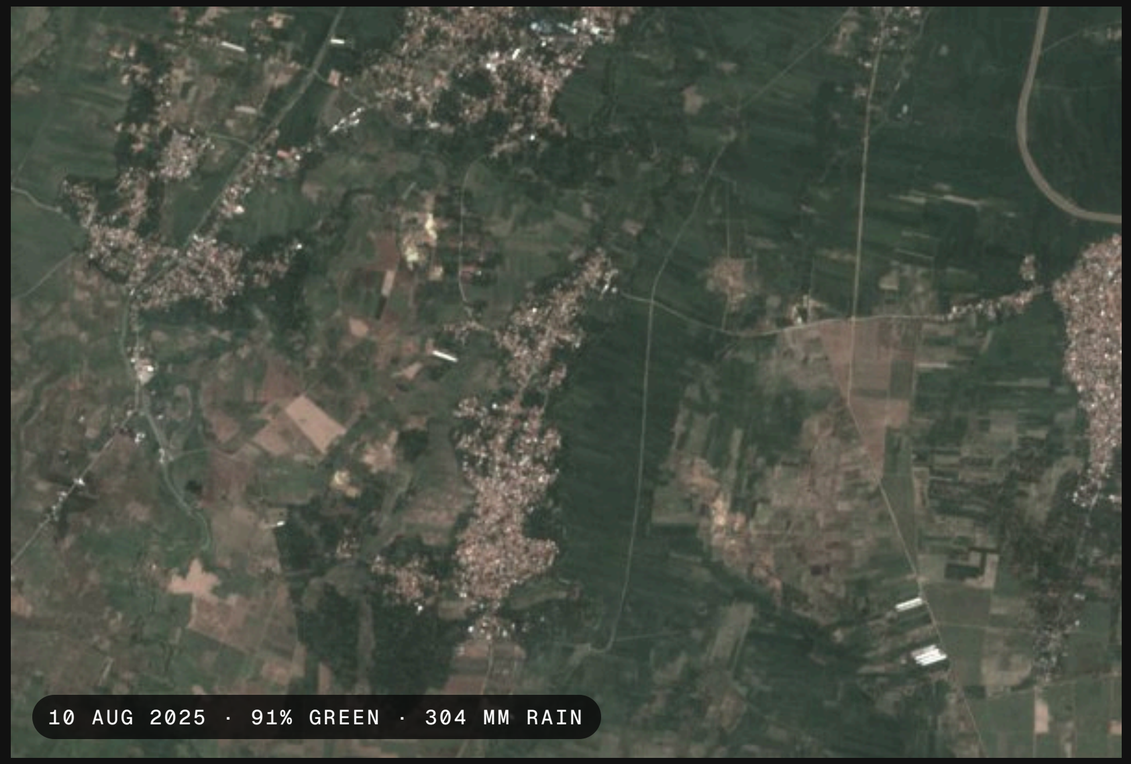
2.6mm
RAIN, 04-20 TO 09-24

534–644mm
IRRIGATION FOR WHEAT

6.6mm/d
PEAK DEMAND

DATA: COPERNICUS SENTINEL-2 L2A (EARTH SEARCH, AWS) · NASA POWER (MERRA-2, FLASHFLUX, IMERG)

West Java: the El Niño, seen from orbit.





DATA: COPERNICUS SENTINEL-2 L2A (EARTH SEARCH, AWS) · NASA POWER (MERRA-2, FLASHFLUX, IMERG) ·
GREEN = MAXIMUM NDVI ABOVE 0.5, JULY–SEPTEMBER

Predicting needs: one plant, one day.

9.7 mm

MAIZE WATER USE PER DAY

ET_0 8.1 mm × Kc 1.20

1.3 L

PER PLANT PER DAY

At 0.13 m² per plant

2.3 kPa

VAPOUR PRESSURE DEFICIT

Drying air

58 mol

DAILY LIGHT INTEGRAL

Estimate from shortwave

EXAMPLE FROM THE NEEDS TOOL, KANSAS SUMMER 2026 MEANS; TOOL VALUES ARE ESTIMATES

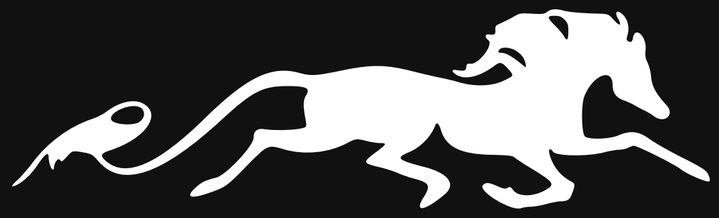
What it costs to be wrong.

PROBE DRIFT	Calibrate EC and pH probes on a schedule	DRIFT IS THE DEFAULT, NOT THE EXCEPTION
PLACEMENT	Shield air sensors; site moisture probes at the roots	SUN ON A SENSOR READS HIGH
OUTAGES	Every controller works on its own	CLOUD-ONLY CONTROL FAILS WHEN THE NETWORK DOES
VISION	A person confirms before anything is sprayed	SCREENING, NOT DIAGNOSIS

FAILURE MODES FROM THE AGRICULTURE-RESEARCH PRIMER; FAIL-SAFES ARE DESIGN RULES

What we build next.

01	A calibration set	PHOTOS OF OUR OWN PLANTS WITH LAB-CONFIRMED CAUSES
02	Sensor logs to the predictor	HORTUSPILOT AND HORTUSROOM READINGS FEED TIER 3
03	Satellite crop coefficients, checked	AGAINST WEIGHING TESTS AND SOIL PROBES ON PILOT BEDS
04	Seasonal outlooks, monthly	STUDY 01’S PIPELINE, RERUN AS THE DATA ARRIVE



What grows.

Analyse a photo, plan a tier and read the paper in the Hortus research pages.